

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Air Handling Clean Care Kit, Part Number 5298776, or equivalent
- Cummins® electronic service tool, or equivalent

Additional Service Items

- Heavy tape
- 5/8 inch deep socket
- Small hammer
- Nylon bristle
- Wire brush
- Pick or scraping tool.

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

Note : Brush away all dirt from around the area of the air handling connections to avoid contamination of the interior of the engine.

- Disconnect the batteries. See equipment manufacturer service information.
- Disconnect the Exhaust Gas Recirculation (EGR) connection tube from the EGR valve. Refer to Procedure 011-070 in Section 11. (</qs3/pubsys2/xml/en/procedures/1016/1016-011-070-tr.html>)
- Disconnect the electrical connection from the EGR valve.

Resistance Check

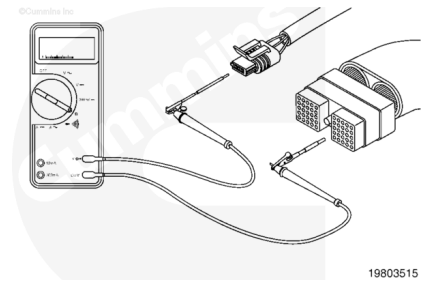
Disconnect the EGR valve motor connector from EGR valve.

Measure the resistance between the EGR valve motor wire A pin and the EGR valve motor wire B pin in the EGR valve motor connector.

Measure the resistance between the EGR valve motor wire A pin and the EGR valve motor wire C pin in the EGR valve motor connector.

Measure the resistance between the EGR valve motor wire C pin and the EGR valve motor wire B pin in the EGR valve motor connector.

The difference between the three resistance values is less than 1 ohm.



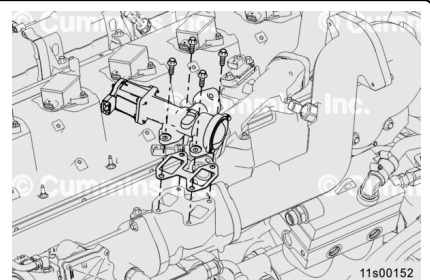
Remove

Clean the area around the EGR valve with a clean cloth to help prevent debris from entering the air intake system when the EGR valve is removed.

Remove the four capscrews holding the EGR valve to the EGR mixer. Remove the EGR valve.

Discard the gasket.

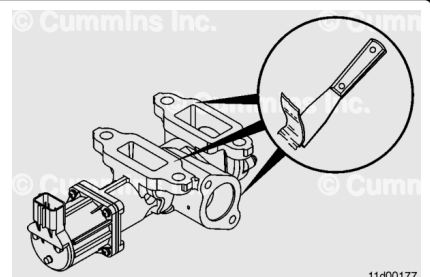
Use protective caps from the Air Handling Clean Care Kit, Part Number 5298776, or equivalent, or heavy tape to cover any open connections to prevent debris from entering the system.



Clean and Inspect for Reuse

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Clean the EGR valve and air intake connection mounting surfaces of gasket material, taking care to keep any debris out of the air intake system.

If the EGR valve has frozen, defrost the sensor using warm air that is below 60°C [140°F].

Dry with compressed air.

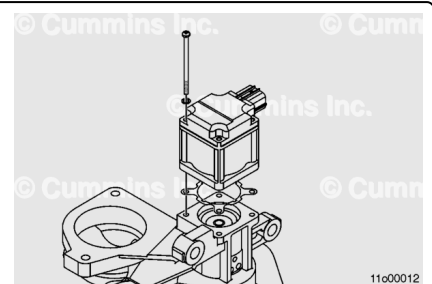
A light coating of carbon is common on the inside of the EGR valve and does **not** need to be removed.

Note : The EGR valve has a green corrosion-resistant coating on its inner surfaces. Do **not** remove this coating.

Remove the four capscrews that hold the EGR valve motor to the EGR valve base.

Remove the EGR valve motor. Take care to keep the shim with the motor.

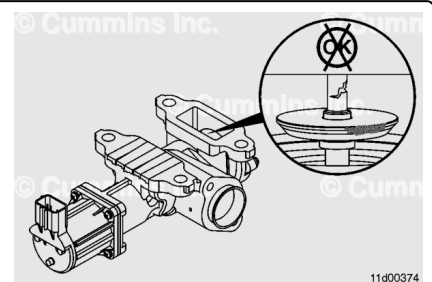
Note : Note the orientation of the electrical connector with respect to the valve body. A witness mark on the outside of the motor case and valve body can be used to document the orientation. This orientation must be the same upon reassembly.



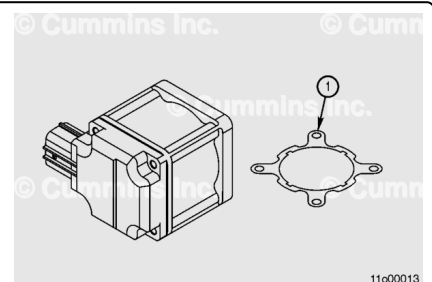
Inspect the EGR valve seats and stem for galling, pitting, cracks, or excessive corrosion.

If the EGR valve shows signs of internal damage, replace the EGR valve.

Note : Soot and condensation leaking from the weep hole on the EGR valve is considered normal and does **not** mean that the EGR valve needs to be replaced.



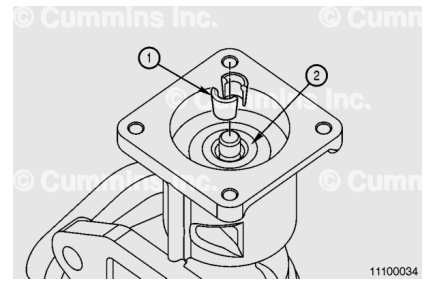
Note : The shim (1) is specific to that valve and can **not** be replaced. If the shim is lost or damaged, replace the EGR valve.



Use two fingers to press down on the valve spring retainer (2) to unlock two valve keepers (1). If valve keepers will **not** release using finger pressure, locate a 5/8 inch deep socket to

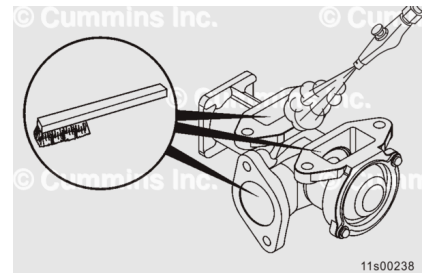
the spring retainer (2). Gently tap on the socket with a small hammer to release the keepers, then use two fingers and press for keeper removal.

Note : The spring, spring keepers, and spring retainer can not be replaced. If the spring, spring keepers, or spring retainer are lost or damaged, replace the EGR valve.

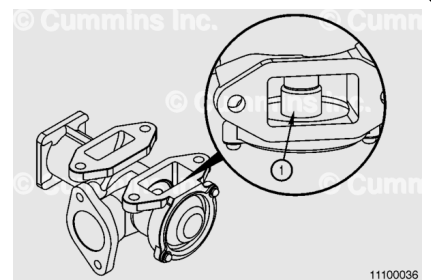


Clean the accumulated soot from the EGR valve inlet port and outlet ports with a nylon bristle and/or wire brush. Some deposits may need to be scraped out with a pick or scraping tool. Remove as much material as possible before applying any cleaner and without damaging the poppet sealing surfaces and the anti-corrosion finish inside the EGR valve body.

Blow the EGR valve inlet and outlet with compressed air to keep any debris out of the EGR valve inside.



Clear the area around the soot guard (1) in the outlet port with a 90 degree bent pick. Clear any deposit material all the way around the front and the back of the soot guard. Make sure that there is no deposit material on or around the soot guard that can hold the poppet shaft open.

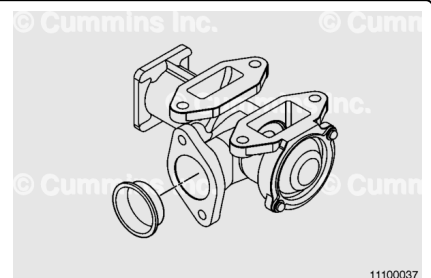


⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

Spray carburetor cleaner into the EGR valve outlet ports to soak inner surfaces. Use plug, Part Number 4919416, available in Air Handling Clean Care Kit, Part Number 5298776, to seal the EGR valve inlet and fill the EGR valve cavity with carburetor cleaner. Soak for 5 minutes. Work the poppet shaft back and forth and turn it while the poppets are in contact with the valve seats (closed position) to help clean the valve. After 5 minutes, drain and properly dispose of the excess cleaner.

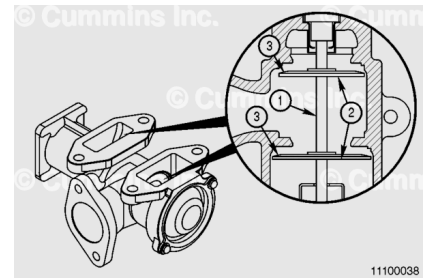
Note : Carburetor cleaner is the most effective solvent for this procedure. Do not substitute any other cleaning solvent.



The EGR valve shaft stem (1) is comprised of two valve faces (3) on two valve seats (2). Use a small nylon or brass wire brush to lightly brush away loose soot from the stem (1), valve seat (3), and also from the valve seat (2) on the EGR valve housing. Take care **not** to damage the poppet sealing surfaces.

Spray additional carburetor cleaner as needed to clean the sealing surfaces.

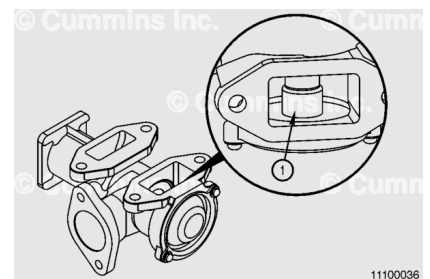
Note : Carburetor cleaner is the most effective solvent for this procedure. Do not substitute any other cleaning solvent.



The soot guard (1) is comprised of outer and inner cylinders, which **must** be able to close with a small gap at the bottom. Use a small nylon or brass wire brush. Brush away deposits from the soot guard. Deposits can be on the back side of the soot guard causing the valve **not** to close fully.

Spray additional carburetor cleaner as needed to clean the sealing surfaces.

Note : Carburetor cleaner is the most effective solvent for this procedure. Do not substitute any other cleaning solvent.

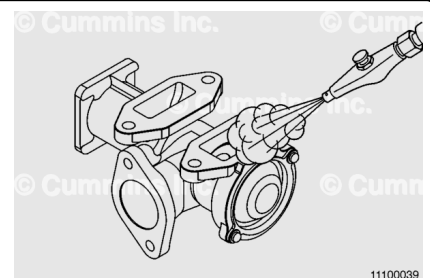


Drain any remaining carburetor cleaner from the EGR valve and properly dispose of the liquid.

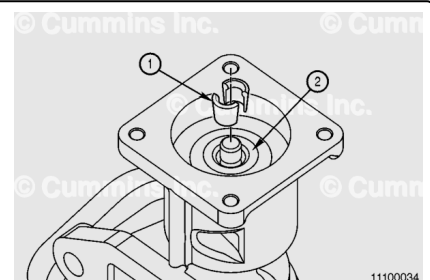
Use compressed air to dry the inside of the EGR valve.

Repeat the cleaning steps, if needed.

Note : An effectively cleaned valve will close with a metal sound when the poppet shaft is closed. A soft thud is an indication that deposits are still holding the EGR valve partially open.



Install the spring, spring retainer (2), and spring keepers (1).

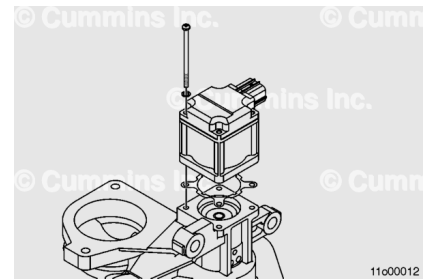


Install the EGR valve motor, shim, and four capscrews to the base.

Torque Value:

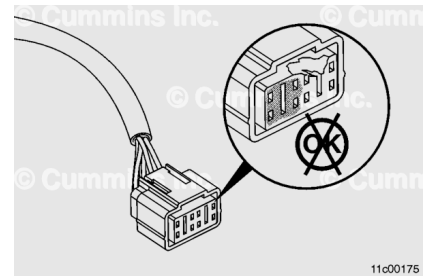
EGR motor capscrews 2.5 n•m [22 in-lb]

Note : EGR valve motor electrical connector must be oriented in the same direction as when originally disassembled.



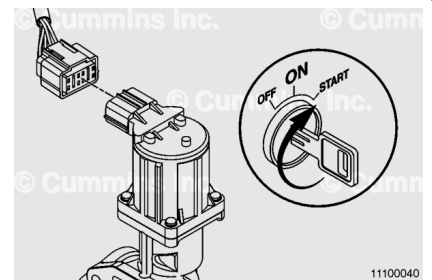
Inspect the engine wiring harness EGR valve actuator connector for corrosion, worn receptacles, or other damage.

Replace the engine wiring harness EGR valve connector if corrosion, worn receptacles, or other damage is found.



⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



Connect the batteries. See equipment manufacturer service information.

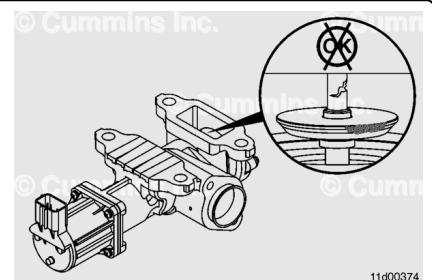
After cleaning the EGR valve, connect the electrical connector to the EGR valve and turn the ignition key ON. Do **not** start the engine. Check for active Fault Code 1896. If Fault Code 1896 persists after cleaning, replace the EGR valve.

Disconnect the batteries. See equipment manufacturer service information.

Inspect the EGR valve seats and stem for galling, pitting, cracks, or excessive corrosion.

If the EGR valve shows signs of internal damage, replace the EGR valve.

Note : Soot and condensation leaking from the weep hole on the EGR valve is considered normal and does not mean that the EGR valve needs to be replaced.

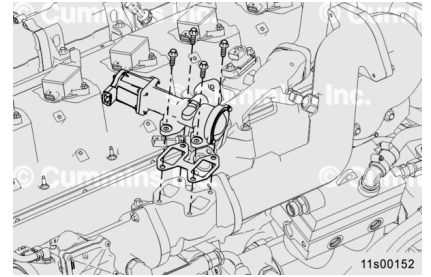


Install

Note : Remove all protective caps from the component.

Install the four capscrews that hold the EGR valve to the EGR mixer with new gasket.

Torque Value: 18 n•m [159 in-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the electrical connection to the EGR valve.
- Connect the EGR connection tube to the EGR valve. Refer to Procedure 011-070 in Section 11. (</qs3/pubsys2/xml/en/procedures/1016/1016-011-070-tr.html>)
- Connect the batteries. See equipment manufacturer service information.
- Use the recommended Cummins® electronic service tool, or equivalent, to reset the fuel tables if EGR valve is replaced. Refer to Procedure 005-263 in Section 5. (</qs3/pubsys2/xml/en/procedures/1016/1016-005-263.html>)
- Operate the engine. Check for leaks.